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Q 1.

(2)	[3]	(4)	5	[6]	7
(8)	[9]	(10)	11	[12]	13
(14)	[15]	(16)	17	[18]	19
(20)	[21]	(22)	23	[24]	25

The unmarked numbers are 5, 7, 11, 13, 17, 19, 23, 25.

All the remaining numbers above are prime numbers except 25

2 and 3 are special prime numbers

Any integer divided by 6 has a remainder in the following:

0, 1, 2, 3, 4, 5

Therefore every integer has one of these 6 forms:

$6q, 6q+1, 6q+2, 6q+3, 6q+4, 6q+5$,

Which means an integer can be written as $6q+r$ where $r \in \{0,1,2,3,4,5\}$

Examine each form.

Form	Remainder.	Reason for elimination or Retainment.
$6q$	0	divisible by both 2 and 3 - composite
$6q+1$	1	possible prime
$6q+2$	2	divisible by 2 - composite
$6q+3$	3	divisible by 3 - composite
$6q+4$	4	divisible by 2 - composite
$6q+5$	5	possible prime

So every integer falls into one of the six residue classes of mod 6 and 4 out of the six are composite. Only $6q+1$ and $6q+5$ are possible primes

let $q = k$.

$\therefore 6q+1 = 6k+1$.

$6q+5 = 6(q+1)-1$

let $q+1 = k$.

$\therefore 6q+5 = 6(q+1)-1 = 6k-1$

$6k+1$ and $6k-1$.

These two equations show that a number greater than 3 is a possible if it is in the form $6k+1$ or $6k-1$

In the list of remaining numbers. 25 is a composite function yet it is in the form $6k \pm 1$

$25 \bmod 6 = 1$ and $17 \bmod 6 = 5$

25 is not prime 17 is prim

This shows that mod 6 test is not a sufficient test for primality.

Also, the mod 6 test is fast since it only costs one modlo operation and one comparison instead of two divisibility tests.